

		$u = \sqrt{2}$
11	B	Using conservation of energy, $KE_X + GPE_X + \text{Energy input} = KE_Y + GPE_Y + \text{Work Done by dissipative forces}$ Taking Y as the reference point, $0 + mgh_X + 0 = \frac{1}{2}mv_Y^2 + 0 + \text{Energy by frictional forces}$ $\text{Energy by frictional forces} = 500(9.81)(30) - \frac{1}{2}(500)(11)^2$ $= 1.2 \times 10^5 \text{ J}$

4

13

D

At the bottom of the circle